

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

Commercial APIs can silently swap to cheaper models; empirical results show software auditing breaks under realistic conditions, while TEEs offer verifiable inference.

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Why substitution happens

- APIs bill for premium models while providers optimize GPU spend.
- Cheaper alternatives can preserve user-perceived quality.
- Black-box interfaces prevent direct verification of served weights.

Substitution types

- 1) Smaller model
- 2) Quantized variant
- 3) Fine-tuned/updated checkpoint
- 4) Different model family

Formal test framing

$H_0: P_{\text{api}}(y|x) = P_{\text{spec}}(y|x)$

$H_1: P_{\text{api}}(y|x) \neq P_{\text{spec}}(y|x)$

Goal: infer whether observed API outputs come from the advertised model or a substitute.

Core challenge

Auditors observe outputs only, not model internals, deployment stack, or routing policies.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Family	MMLU	GSM8K	MATH	GPOA Diamond
Llama-3-8B	62.69 → 62.43	61.14 → 60.90	20.65 → 14.91	22.62 → 20.14
Llama-3-70B	78.05 → 77.88	88.06 → 87.35	35.69 → 35.75	29.60 → 33.12
Gemma-2-9B	71.86 → 71.92	81.80 → 79.41	33.41 → 32.53	28.64 → 27.81
Qwen2-72B	82.18 → 81.98	86.72 → 86.82	37.39 → 37.67	29.93 → 31.08
Mistral-7B	59.15 → 58.77	35.90 → 32.20	8.94 → 7.68	21.60 → 22.72

Most MMLU/GSM8K changes are small relative to reported variance.

Some tasks improve post-quantization, shrinking deterministic signatures.

Implication: benchmark-score auditing has weak separation power.

Higher Temperature Amplifies Performance Gaps but Increases Noise

 Benchmark accuracy vs temperature

Across temperatures 0.2, 0.5, 1.0, 2.0, higher sampling temperature degrades benchmark accuracy for all model families.

At high temperature, substitution signal is confounded by broad stochastic degradation, making reliable attribution harder.